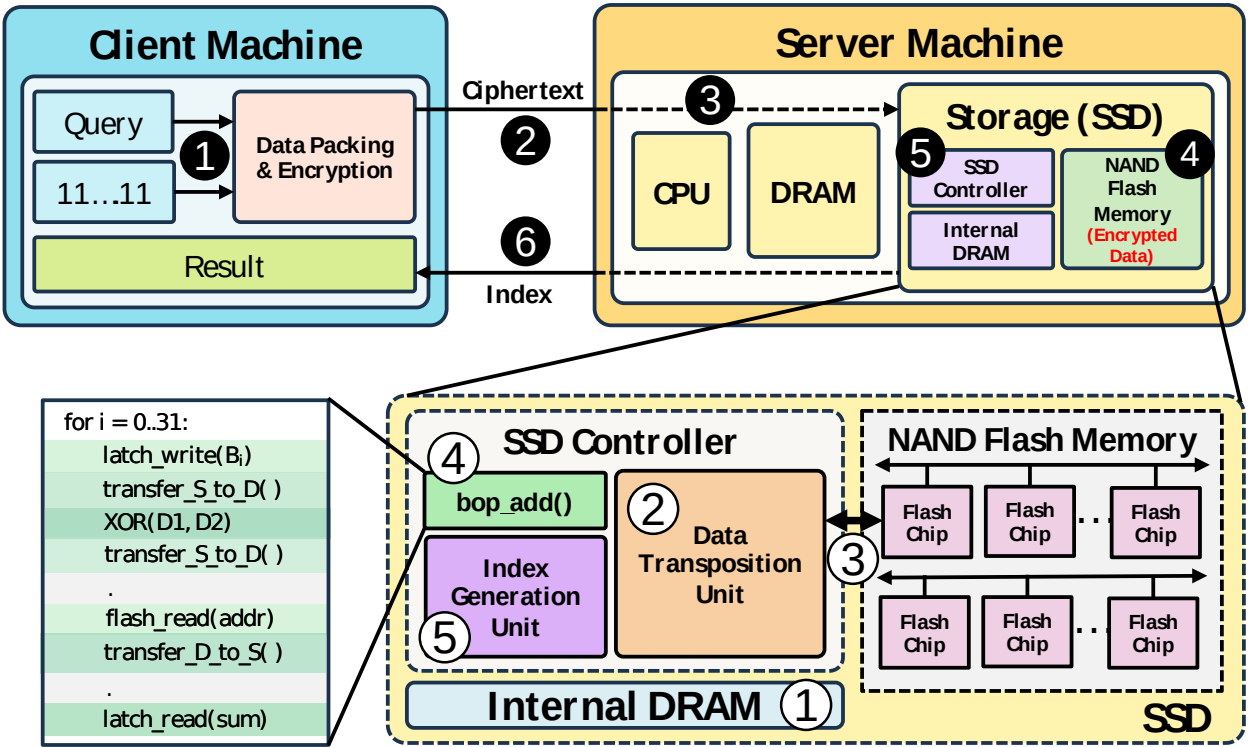
Client Machine
Query
11…11
1
Data Packing & Encryption
Result
Ciphertext
2
6
Index
Server Machine
3
CPU
DRAM
Storage (SSD)
5
SSD Controller
Internal DRAM
NAND Flash Memory (Encrypted Data)
4
for i = 0..31:
latch_write(Bi)
transfer_S_to_D( )
XOR(D1, D2)
transfer_S_to_D( )
.
flash_read(addr)
transfer_D_to_S( )
.
latch_read(sum)
SSD Controller
4
bop_add()
Index Generation Unit
5
2
Data Transposition Unit
3
NAND Flash Memory
Flash Chip
Flash Chip
Flash Chip
Flash Chip
Flash Chip
Flash Chip
Internal DRAM
1
SSD